Hilbert Scheme

Joseph Sullivan

September 2025

Contents

1 Some Moduli Functors and Families

1.1 Technicalities

We are going to be defining some moduli functors of interest, but in order to define our notions of families, we will need some technical definitions.

Definition 1.1. A ring map $R \to S$ is of **finite presentation** if there exist finitely many R-algebra generators for S with finitely many relations, so

$$S \cong R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

and the map from R is the inclusion to the polynomial ring (and the projection).

Definition 1.2. Let $X \xrightarrow{f} S$.

- We say f is of **finite presentation** at $x \in X$ if there exists an affine open neighborhood $\operatorname{Spec}(A) = U \subseteq X$ of x and affine open $\operatorname{Spec}(R) = V \subseteq S$ with $f(U) \subseteq V$ such that the induced ring map $R \to A$ is of finite presentation.
- Then, f is **locally of finite presentation** if it is of finite presentation at every point of X (so this is local on the base).
- Finally, f is **finite presentation** is is it locally of finite presentation, quasi-compact, and quasi-separated.

Example 1.3. When S is locally Noetherian, locally of finite type already implies locally of finite presentation, since the kernel of the map on sections will be finitely generated by the Noetherian assumption.

About morphisms of finite presentation, we have the following fact (which we will not prove).

Theorem 1.4. A morphism $f : X \to S$ is of finite presentation if and only if for all directed systems of affine schemes T_i,

$$\operatorname{colim} \operatorname{Hom}_S(T_i, X) = \operatorname{Hom}_S(\lim T_i, X).$$

(Note that we are taking in a colimit rather than a limit as is usual, and the limit ends up on the left argument, also unusual—this looks something like the definition of a compact object in a category).

Definition 1.5. Let $X \xrightarrow{f} S$, and $\mathcal{F}$ a sheaf on X.

We say $\mathcal{F}$ is **flat** over S at a point $x \in X$ if the stalk $\mathcal{F}_x$ is a flat $\mathcal{O}_{S,f(x)}$-module (by extension of scalars).

We say $\mathcal{F}$ is **flat** over S if it is flat over S at every point.

We say f is **flat** if $\mathcal{O}_X$ is flat over S.

1.2 The Functors

We first give a warm-up moduli functor.

Definition 1.6. Let V be a vector bundle of rank n on a Noetherian scheme S, and q an integer with $0 < q < n$. Define the functor

$$\begin{aligned} \operatorname{Gr}(q, V) : \mathrm{Sch}^{\mathrm{op}} &\to \mathrm{Set} \\ (T \to S) &\mapsto \{\text{vector bundle quotients } V_T \twoheadrightarrow Q \text{ on } T \text{ of rank } q\}. \end{aligned}$$

We will call this the **Grassmannian**, especially after we establish that it is representable by a projective scheme over S. In the special case that $S = \operatorname{Spec} k$ (for k a field or maybe $\mathbb{Z}$), then V is a vector space (or an abelian group), and we recover the classical Grassmannian.

Definition 1.7. Let S be a Noetherian scheme, F a coherent sheaf on $\mathbb{P}^n_S$ that can be written as a quotient of $\mathcal{O}_{\mathbb{P}^n_S}(-\ell)^{\oplus r}$ for some l, r. Finally, fix a polynomial $P \in \mathbb{Q}[z]$. Define the functor

$$\begin{aligned} \operatorname{Quot}^P(F/\mathbb{P}^n_S/S) : (\mathrm{Sch}/S)^{\mathrm{op}} &\to \mathrm{Set} \\ (T \to S) &\mapsto \left\{ \begin{matrix} \text{quasi-coherent and finitely presented quotients} \\ F_T \twoheadrightarrow Q \text{ on } \mathbb{P}^n_T \text{ such that } Q \text{ is flat over } T \text{ and} \\ Q_t := Q|_{\mathbb{P}^n_{\kappa(t)}} \text{ has Hilbert polynomial } P \text{ for all } t \in T \end{matrix} \right\} \end{aligned}$$

which we will call the **Quot Scheme** (once we know it is represented by a scheme projective over S).

In words, $\mathrm{Quot}^P(F/\mathbb{P}^n_S/S)$ is the moduli functor for quotients Q of F with a fixed Hilbert polynomial P. The condition that F can be written as a quotient $\mathcal{O}_{\mathbb{P}^n_S}(-\ell)^{\oplus r}$ is automatically satisfied for $S = \mathrm{Spec}\, A$ for a Noetherian ring A, by Serre vanishing (so this should be thought of as a technical condition necessary for the relative case).

We can then specialize to the setting where $F = \mathcal{O}_{\mathbb{P}^n_S}$, so that quotients are exactly the (pushforwards of) structure sheaves of closed subschemes (i.e., those cut out by the kernel of the quotient map, the ideal sheaf). Then, F_T is just $\mathcal{O}_{\mathbb{P}^n_T}$, so Q is a quotient $\mathcal{O}_Z$ for a closed subscheme

$$Z \subseteq \mathbb{P}^n_T.$$

The condition that $\mathcal{O}_Z$ is flat over T is exactly what it means for the composition $Z \to \mathbb{P}^n_S \to S$ to be flat. Since Z is a closed subscheme of projective space,

$$Z \to \mathbb{P}^n_S \to S$$

is separated (so in particular quasi-separated), and since S is Noetherian, we similarly get Z is quasi-compact. All this is to show that $Z \to S$ is finitely presented, where the finite presentation

$$\mathcal{O}_{\mathbb{P}^n_T} \to \mathcal{O}_Z$$

will give us a finitely type ideal sheaf, which will give us the finitely many algebra generators for the local model of the map $Z \to S$.

All this being said, we have the following definition.

Definition 1.8. Let S be a Noetherian scheme and $P \in \mathbb{Q}[z]$. Define the functor

$$\begin{aligned} \mathrm{Hilb}^P(\mathbb{P}^n_S/S) : (\mathrm{Sch}/S)^{\mathrm{op}} &\to \mathrm{Set} \\ (T \to S) &\mapsto \left\{ \begin{array}{l} \text{closed subschemes } Z \subseteq \mathbb{P}^n_T \text{ flat and finitely presented over } T \\ \text{such that } Z_t \subseteq \mathbb{P}^n_{\kappa(t)} \text{ has Hilbert polynomial } P \text{ for all } t \in T \end{array} \right\} \end{aligned}$$

which we will call the **Hilbert scheme** once we know it is represented by a scheme projective over S.

2 Some Theory

We want to think of all contravariant functors $h : (\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Set}$ as spaces, like $h_X = \mathrm{Hom}(-, X)$. One immediate necessary condition for such a contravariant functor to be representable is it being a *Zariski sheaf.*

Definition 2.1. A functor $h : ()\mathrm{Sch}/S)^{\mathrm{op}} \to \mathit{Set}$ is called a **Zariski sheaf** if given an open cover

$$X = \bigcup_i U_i,$$

we get an equalizer sequence

$$\bullet \to h(X) \to \prod_i h(U_i) \rightrightarrows \prod_{i,j} h(U_i \cap U_j).$$

We will then carry over properties originally defined for morphisms of schemes to natural transformations $h' \to h$ by saying $h' \to h$ has property P whenever base changed along a scheme X, we get a morphism of schemes with property P.

That is, given a natural transformation $h_X \to h$ (where $h_x = \mathrm{Hom}(-, X)$), we can then form the pullback

$$\begin{tikzcd} h_X \times_h h' \arrow[r] \arrow[d] & h_X \arrow[d] \\ h' \arrow[r] & h \end{tikzcd}$$

(where the pullback of functors is by pulling back in the codomain category, Set, in each object), and then we ask that $h_X \times_h h'$ is representable by a scheme Y, and the morphism $h_Y \to h_X$ corresponds to a morphism

$$Y \to X$$

with property P.

Definition 2.2.
- For two functors $h', h : (\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Set}$, we say a natural transformation $h' \to h$ expresses h' as a **subfunctor** of h if each component $h'(X) \to h(X)$ is injective.
- We say $h' \hookrightarrow h$ is an **open subfunctor** if it base changes to open immersions, i.e., given $h_X \to h$, $h_X \times_h h' \cong h_U$ for some scheme U, and the morphism $h_U \to h_X$ is an open immersion.
- We say a collection h_i of open subfunctors **covers** h if it base changes to open covers, i.e., given $h_X \to h$, for the $U_i \hookrightarrow X$ by

  $$h_{U_i} \cong h_X \times_h h_i \hookrightarrow h,$$

  the U_i form an open cover of X.

Proposition 2.3. A locally representable (having an open cover by representable functors) Zariski sheaf is representable.

Proof. Glue the open subfunctors h_{U_i} along their inclusion into h, i.e., we can define $h_{U_{ij}}$ as the fiber product

$$h_{U_i} \times_h h_{U_j}$$

and glue the U_i along the U_{ij}. This gives us a scheme X.

Next, build a morphism $\alpha : h_X \to h$, by seeing that this is the data (by Yoneda) of an $\alpha \in h(X)$, which we get exactly by the Zariski sheaf property, since α will push to the tuple of inclusions $\iota_i \in h(U_i)$, which are cooked up to agree on overlaps.

Finally, check that α is an isomorphism. From its construction, it isn't hard to see that it is an isomorphism when base changed to each U_i. Then, to check when base changing to an arbitrary T, finish this!!

by seeing injective and surjective. Pick a test scheme T, so that we will study the set function $h_X(T) \to h(T)$.

First, show surjective. Pick $\beta \in h(T)$ to try to reach. This is (by Yoneda) the data of $\beta : h_T \to h$. Base change this along X, and we want to argue that $h_X \times_h h_T \cong h_T$ via the projection, so that our pullback diagram is

$$\begin{tikzcd} h_T \arrow[r] \arrow[d, "="] & h_X \arrow[d] \\ h_T \arrow[r] & h \end{tikzcd}$$

which factors $h_T \to h$ through h_X, providing a preimage of β in $h_X(T)$.

So, to see $h_X \times_h h_T \cong h_X$, we che

To do this, recognize that by Yoneda, $\iota : h_U \to h$ is the data of an element $\iota \in h(U)$, and then $\square$

This then gives us a strategy to show certain moduli functors are representable—find an open cover (after you prove it is a Zariski sheaf).

3 Grassmannian

TODO: prove the Grassmannian is projective by Plucker embedding. Do the relative version. Also prove projective by proving proper and Chow's lemma lemma???